

ST. PETERS HIGH SCHOOL BUSABALA

COMPETENCE BASED ASSESSMENT

END OF TERM I EXAMS 2026

UGANDA ADVANCED CERTIFICATE OF EDUCATION (UACE)

S.6 BIOLOGY – PAPER 1 (THEORY)

Time: 3 Hours

NAME..... SIGN.....

Instructions

- Answer **four items only**
- **Section A is compulsory**
- In **Section B**, answer **one item from each part**

SECTION A (COMPULSORY)

Item 1

A biogas company is using **organic waste (animal dung and food waste)** to produce methane gas through microbial activity in sealed digesters. However, some digesters produce **less gas than expected**.

An investigation compared different conditions

Digester condition	Oxygen (%)	Methane production (L/day)	Temperature (°C)	Microbial activity (units)
Well-sealed, warm	2	45	35	90
Poorly sealed, warm	10	20	34	50
Well-sealed, cool	2	15	20	40
Poorly sealed, cool	12	8	20	25

Task

- a) Analyse the relationship between **oxygen levels, temperature, and methane production**. (5 marks)

- b) Explain the **biological process responsible for methane production in the digesters**. (6 marks)

- c) Using the data, explain why **oxygen reduces methane production**. (6 marks)

d) Advise the company on the **optimal conditions for maximum gas production**, giving reasons. (5 marks)

e) Predict the effect of increasing temperature beyond 40°C on microbial activity. (3 marks)

Item 2

A company is growing crops in a **hydroponic system (plants grown without soil)**. After some time, plants showed **reduced growth despite sufficient water supply**.

The nutrient solution was analysed:

Nutrient concentration (%)	Water uptake (ml/day)	Leaf turgidity (index)	Growth rate (cm/week)
0.5	60	85	5.2
1.0	50	75	4.5
2.0	30	55	2.5
3.0	15	30	1.0

Task

a) Analyse the relationship between **nutrient concentration and plant growth**. (5 marks)

b) Explain how **high solute concentration affects water uptake in plant cells**. (6 marks)

c) Using the data, explain why **leaf turgidity decreases at higher concentrations**. (6 marks)

d) Suggest why plants begin to **wilt even when water is available**. (4 marks)

e) Propose **three ways to optimise plant growth in hydroponic systems**. (4 marks)

SECTION B

Part I (choose one item)

Item 3

A mining company operates in a region with **limited access to clean drinking water**. Workers often consume contaminated water and show symptoms of **frequent urination, dehydration, and electrolyte imbalance**.

A health study recorded the following

Worker	Water quality (index)	Urine output (L/day)	Body water loss (L/day)	Blood ion concentration (mmol/L)
A	Poor	3.0	2.5	160
B	Moderate	2.5	2.0	150
C	Good	1.8	1.5	140
D	Very good	1.5	1.2	135

Task

- a) Analyse the relationship between **water quality, urine output and ion concentration**. (5 marks)
- b) Explain how **dehydration affects the balance of body fluids and ions**. (6 marks)
- c) Describe how the body regulates **water and salt balance under normal conditions**. (6 marks)
- d) As a public health officer, recommend **three interventions to improve worker health**, with justification. (4 marks)
- e) Predict the long-term effects of continued electrolyte imbalance. (4 marks)

item 4

A transport company monitored drivers working long hours in **high-altitude mountainous regions** where oxygen levels are lower than normal.

The following data was collected:

Altitude (m)	Oxygen availability (%)	Breathing rate (breaths/min)	Heart rate (beats/min)
0	21	12	70
1500	17	18	85
3000	14	25	100
4500	11	32	120

Task

- Analyse the relationship between **altitude and physiological responses**. (5 marks)
- Explain why **breathing rate and heart rate increase at higher altitudes**. (6 marks)
- Explain how the body attempts to **maintain oxygen supply under low oxygen conditions**. (6 marks)
- Suggest **three measures drivers can take to reduce altitude-related health risks**. (4 marks)
- Evaluate the limits of the body's ability to adapt to extreme altitude. (4 marks)

Part 11 (choose one item)

Item 5

A hospital introduced a new sterilization method to control bacterial infections. Initially effective, but over time some bacteria survived.

Time (months)	% bacteria surviving
1	3
3	10
6	22
9	40
12	65

Task

- Analyse the trend in bacterial survival. (4 marks)
- Explain how **genetic variation leads to survival of some bacteria**. (7 marks)
- Predict what will happen if the same sterilization method continues to be used. (4 marks)
- Propose **four alternative infection control strategies**, with biological reasoning. (6 marks)
- Evaluate the implications for **hospital infection control systems**. (4 marks)

Item 6

A landfill site near a town has been accumulating waste over several years. Environmental monitoring showed the following:

Distance from landfill (m)	Plant growth (index)	Soil oxygen (%)	Decomposer activity (index)	Pest population (index)
10	20	8	90	120
50	40	12	70	90
100	60	16	50	60
200	80	20	30	30

Task

- a) Analyse the relationship between **distance from landfill and environmental conditions**. (5 marks)
- b) Explain how **decomposition processes affect soil oxygen levels**. (6 marks)
- c) Account for the increase in pest population near the landfill. (5 marks)
- d) Propose **four strategies to manage the landfill sustainably**. (6 marks)
- e) Evaluate the impact of landfill sites on **ecosystem health**. (3 marks)

END

“Whether you think you can or think you can't, you're right.”